

Lesson 5 MoveIt Kinematic Model and Motion Planning

1. Preface

1.1 kinematic model introduction

Kinematics is a branch of mechanics that describes and studies the variation law of the object position over time based on geometry, which does not involve the physical properties of the object itself and the force applied on the object. The so-called kinematic model is a mechanical tool for studying the laws of object position and time change.

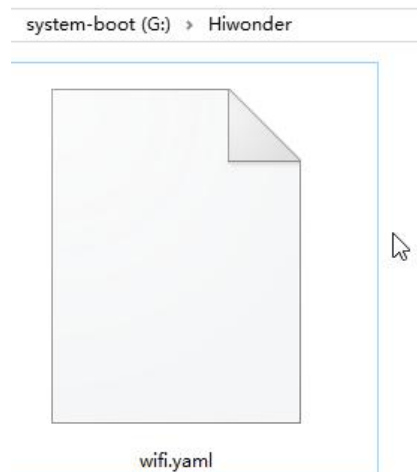
1.2 Motion Planning Introduction

Motion Planning is to find the path meeting the constraints for the object locating between position A and position B. This constraint can be collision free, shortest path, minimum mechanical work, etc. For example, plan a collision-free route for robotic arm to realize object picking.

2. Set Up Environment

2.1 LAN Configuration

1) Insert the SD card into the card reader and connect it to the USB port. Then open the image file to find the Wi-Fi configuration file “Wi-Fi.yaml” and open it with Notepad.

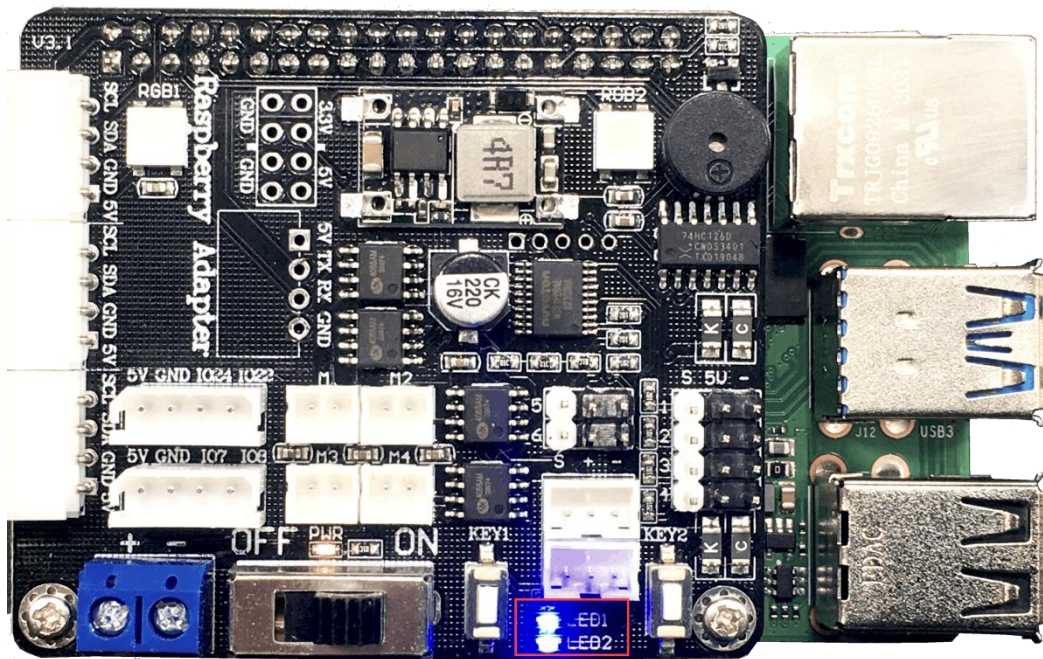


2) Modify the content according to the figure below and save.

```
1  #mode: "client"
2  #ap_mode:
3  #  ssid: "hiwonder"
4  #  band: 5
5  #  channel: 149
6  #  password: "123456789"
7  #  gateway: "192.168.149.1"
8  client_mode: ①
9      ssid: "LH_5G" ②
10     password: "lehuansuoer3" ③
11  #  timeout: 30
12
```

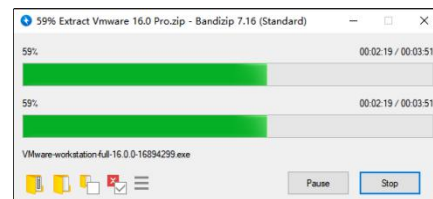
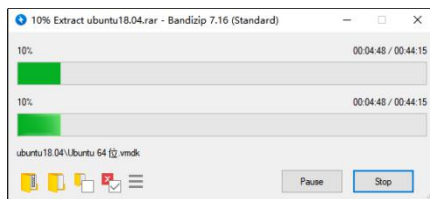
Note: ① refers to LAN mode, ② represents Wi-Fi name and ③ is for Wi-Fi password. And you should input the name and password of your own Wi-Fi and the the picture is for example only.

3) Then insert the image SD card back to the Raspberry Pi controller, restart the robotic arm and wait until the network is distributed successfully. When the LED1 and LED2 on the controller light up, it means the configuration succeeds.

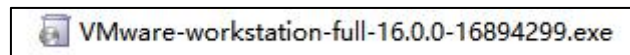


2.2 Set up virtual environment

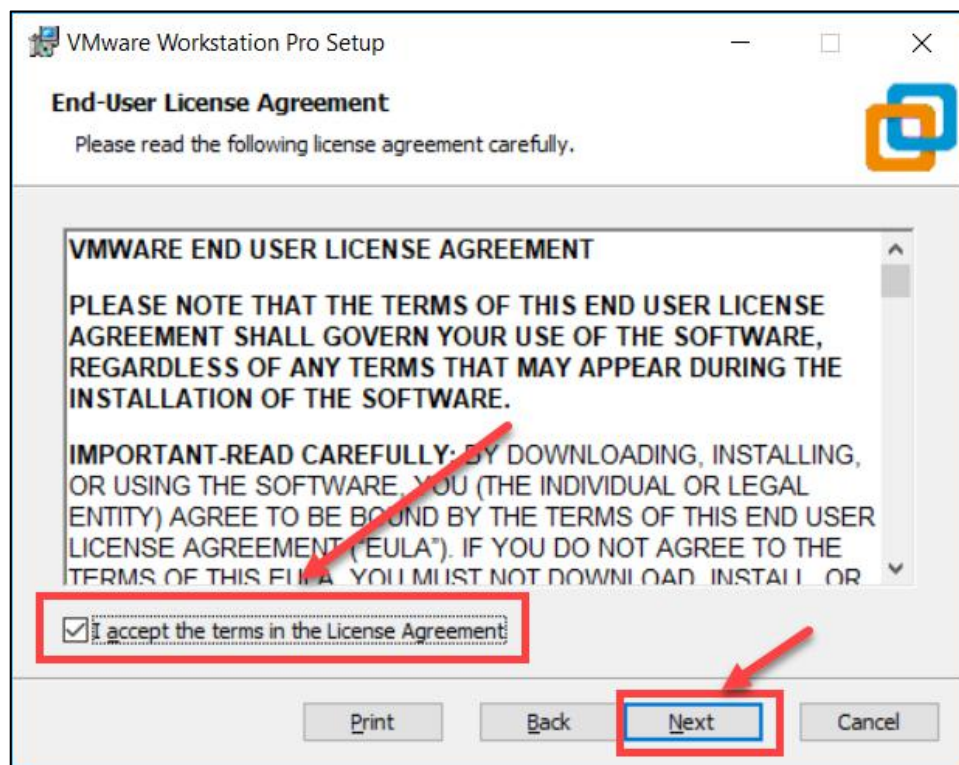
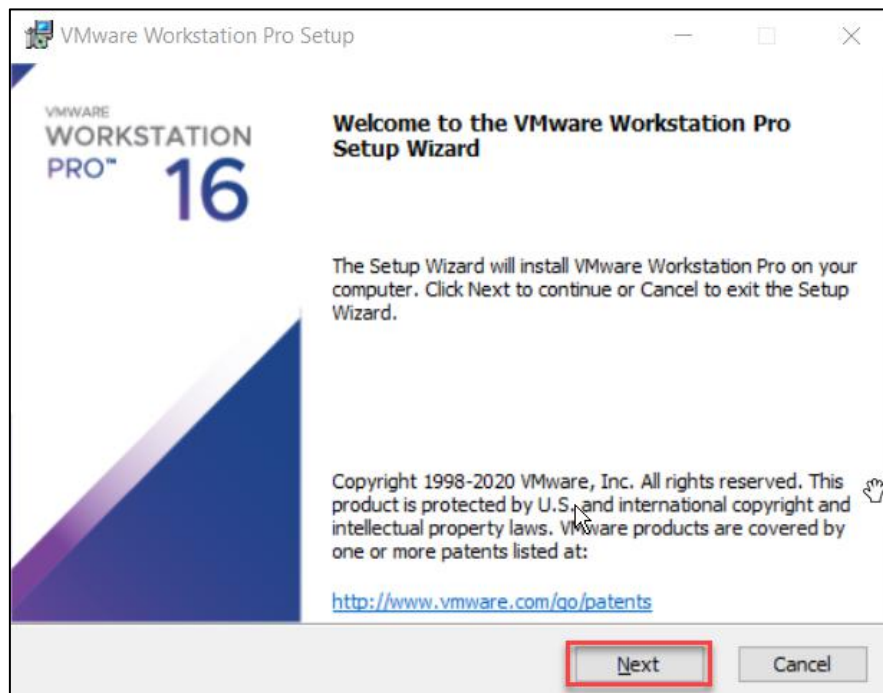
1) Decompress the Ubuntu image package and virtual machine package in the same directory.

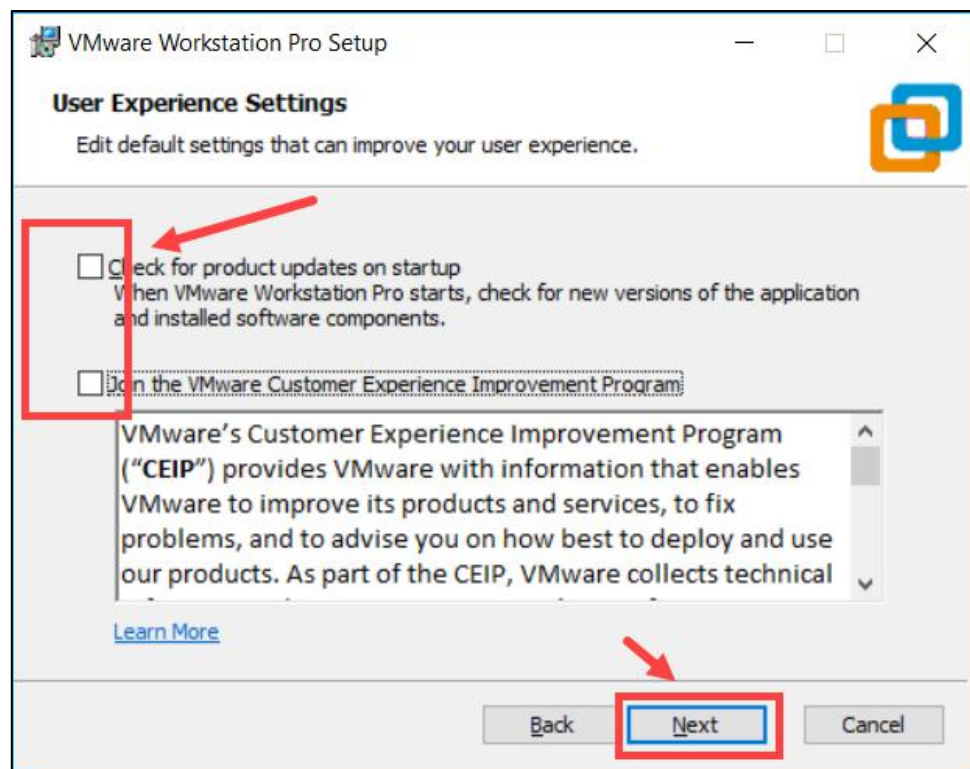
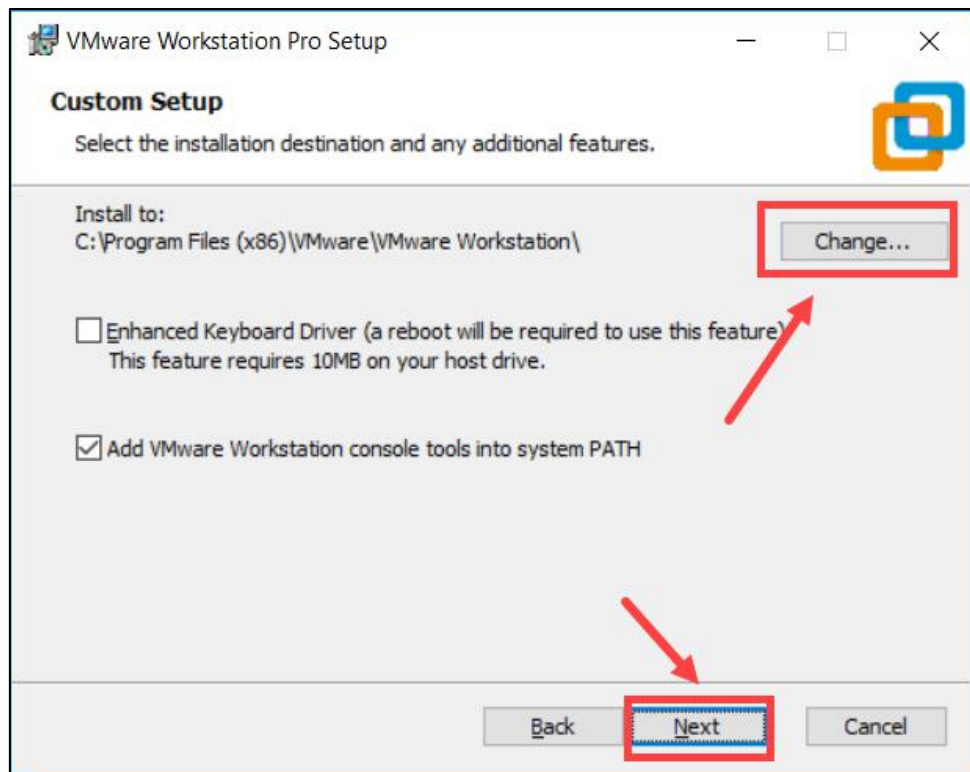


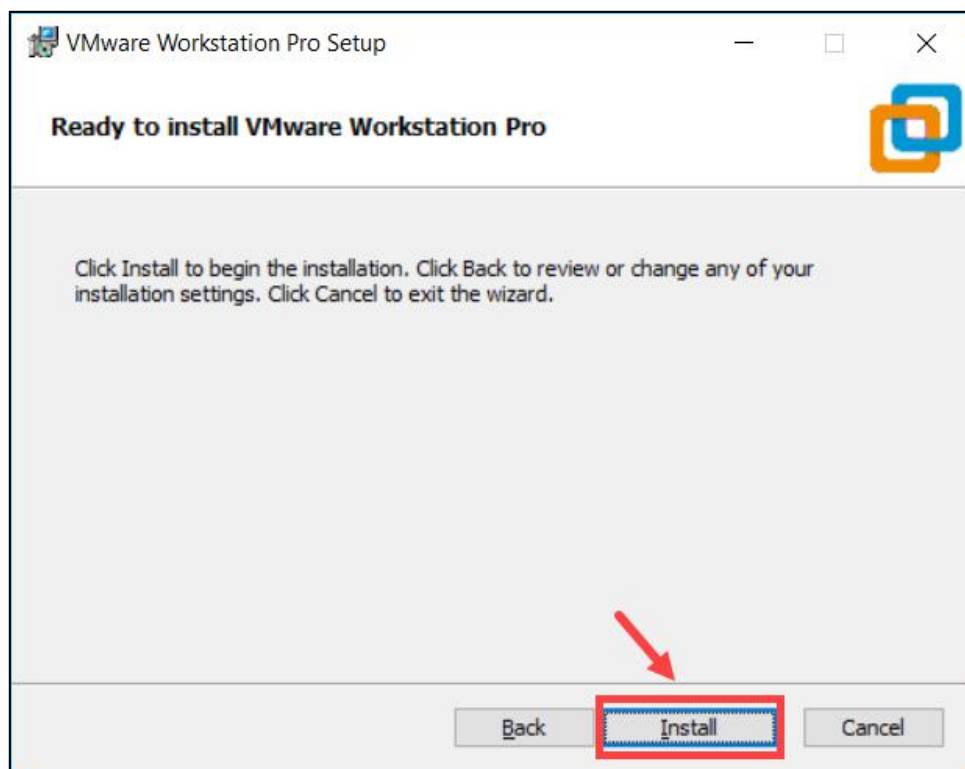
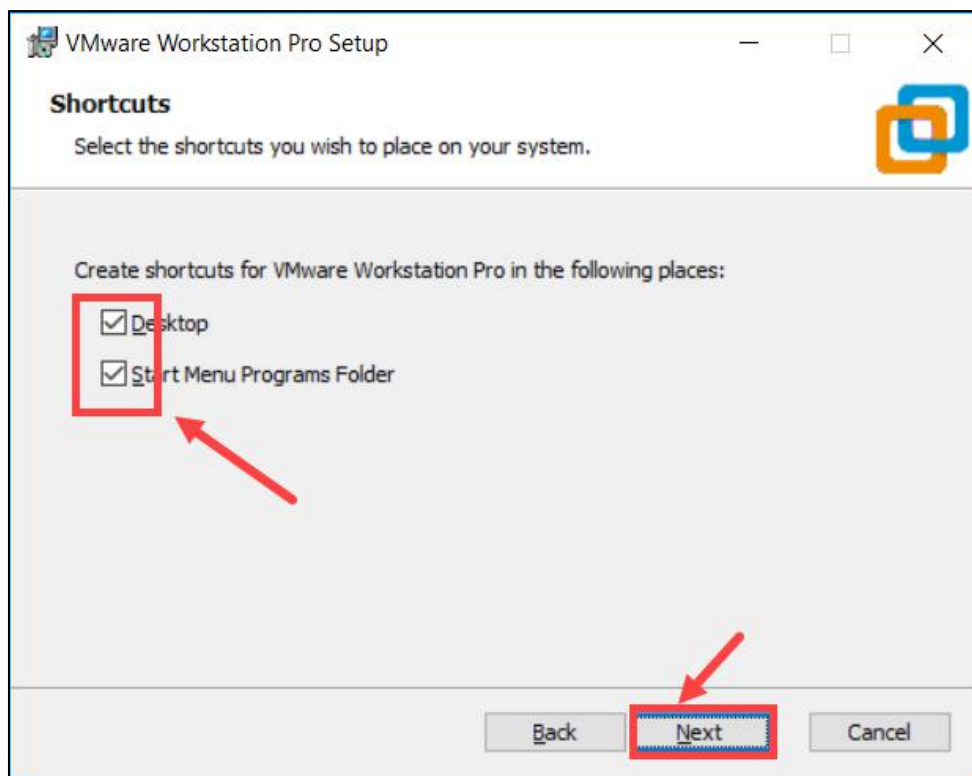
2) Then locate the decompressed virtual machine folder and double-click the virtual machine executable file with extension .exe.

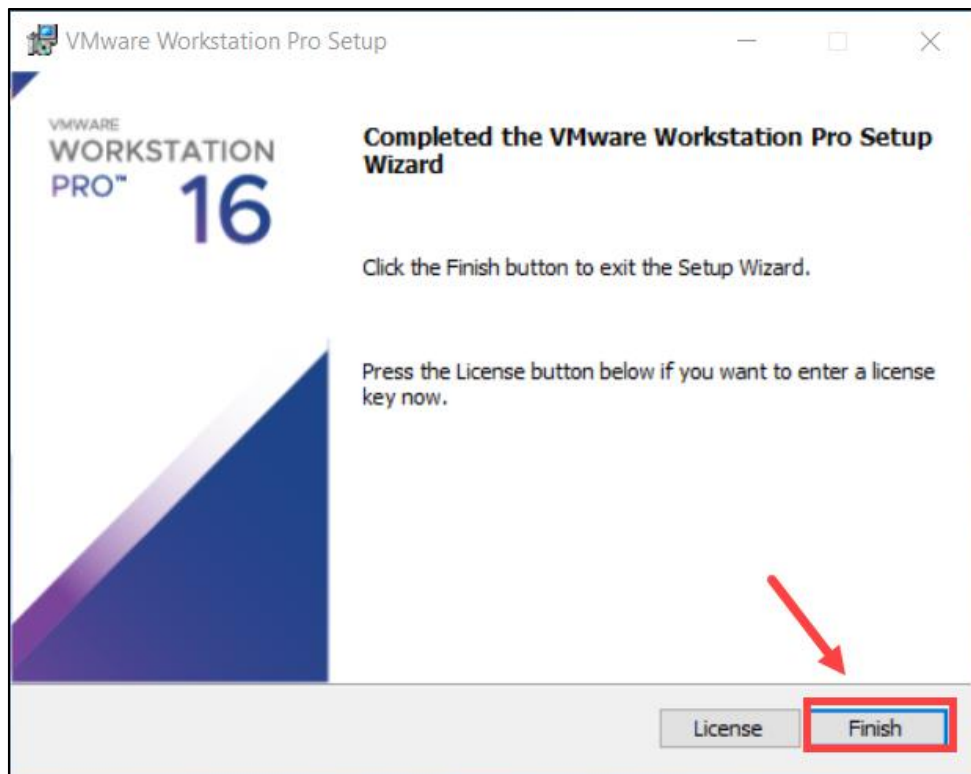


3) Then complete the installation of virtual machine following the picture order.





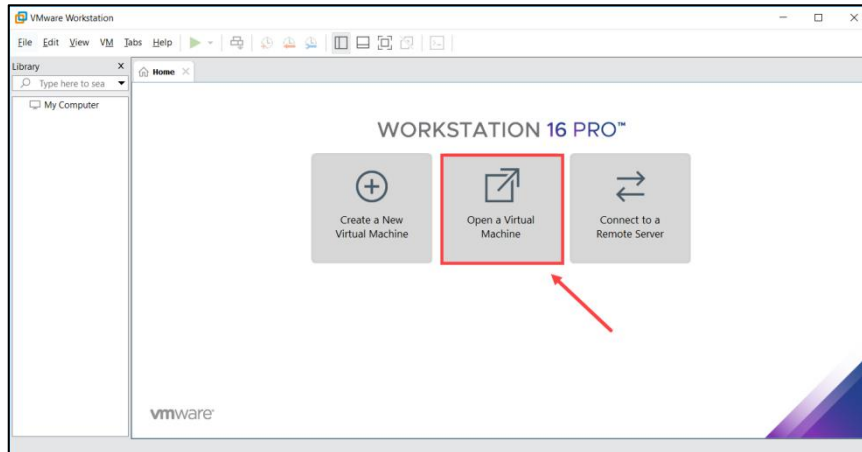




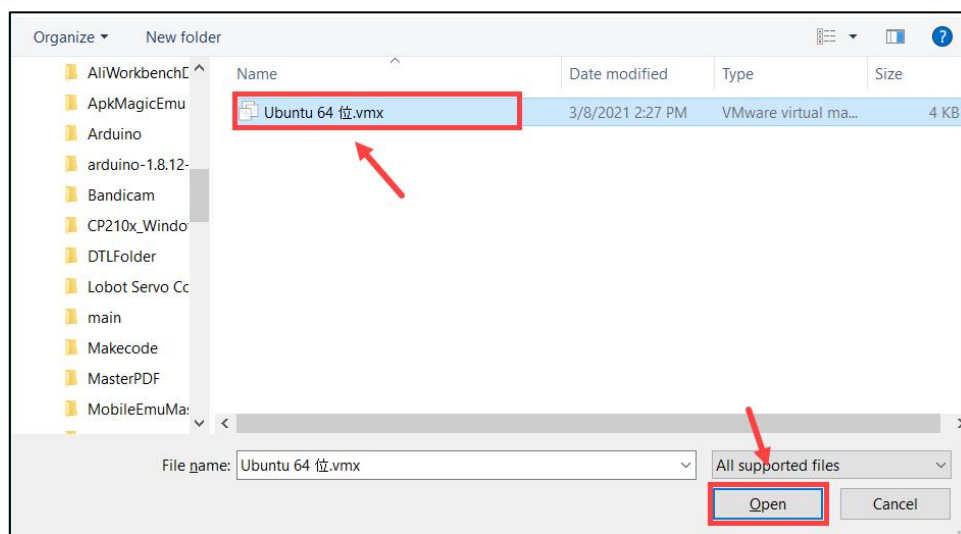
4) It is required to enter the key for the first use of virtual machine. And you can enter any key offered by “VMware16 Activation Key” folder, then click “continue”.



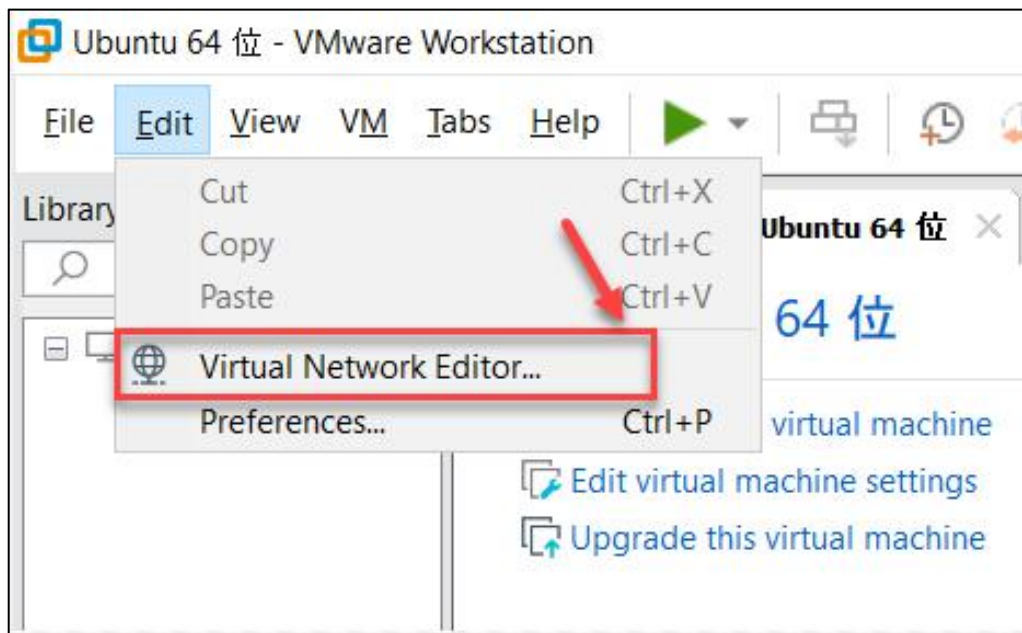
5) After entering the virtual machine interface, select “Open a Virtual Machine”.



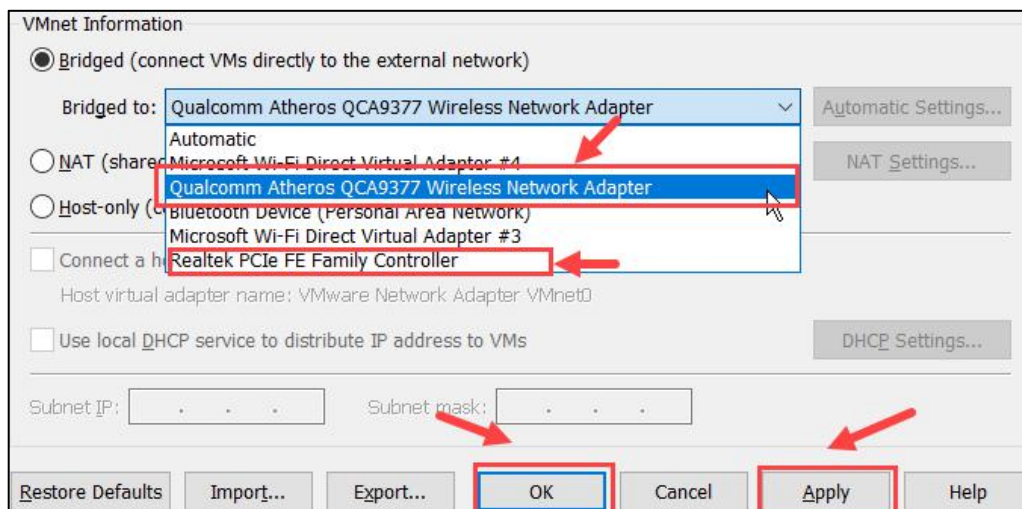
6) When the window pops up, select “Ubuntu64 位.vmx” locating in the decompressed ubuntu image and open.



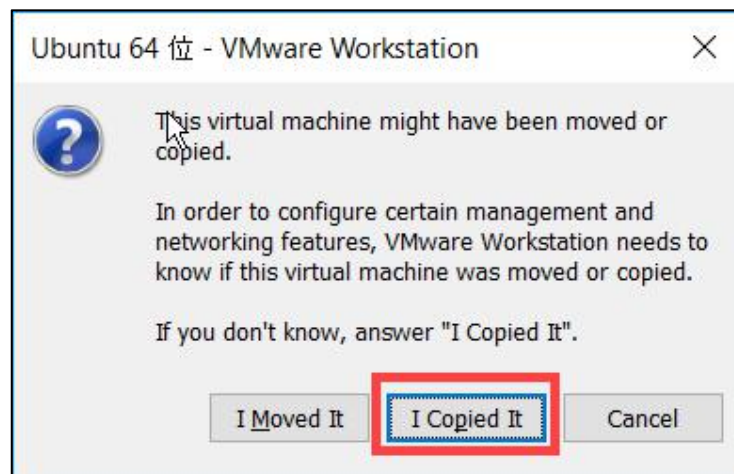
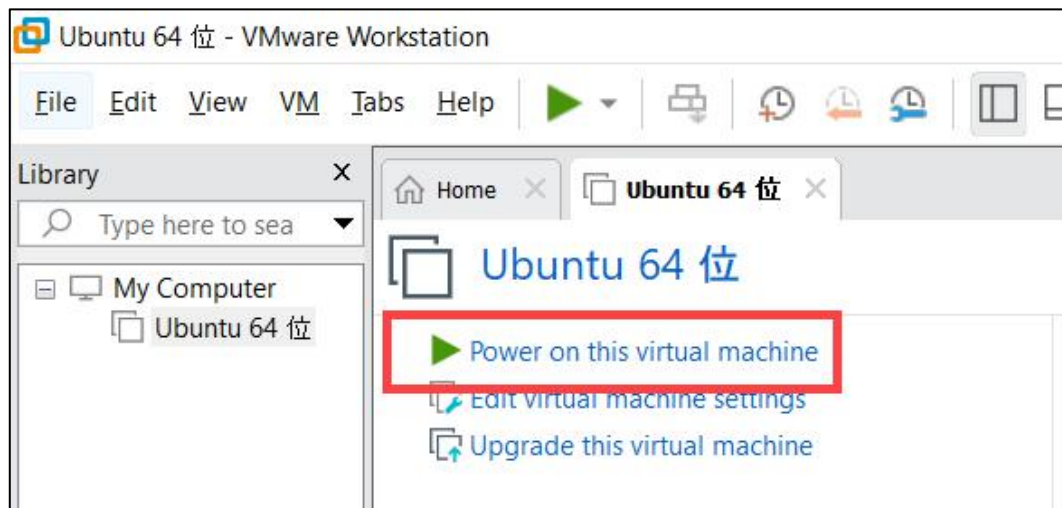
7) Before starting the virtual machine, please configure the network. Click “Edit” and select “Virtual Network Editor”.



8) Select the appropriate Bridge mode based on the type of network interface card, next select “Apply” and click “OK”.

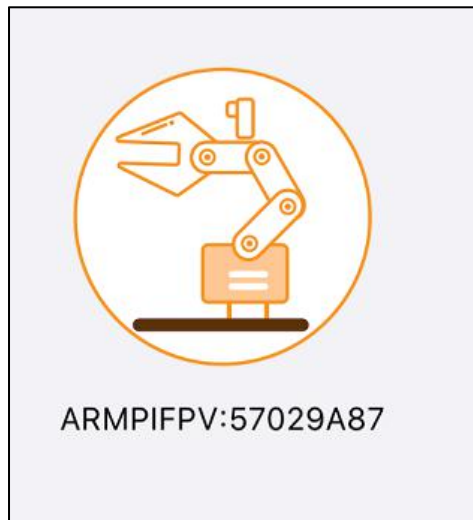


9) Click “Power on this virtual machine” after the above steps. And select “I Copied it” in the pop-up window.

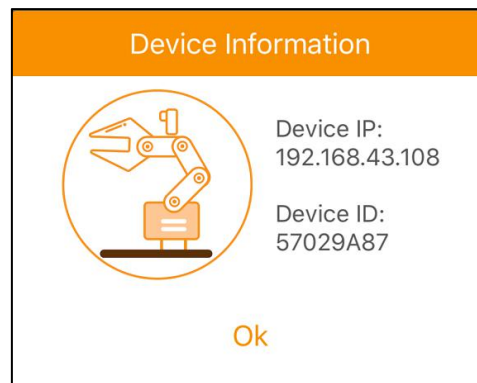


2.3 Development Environment LAN Connection

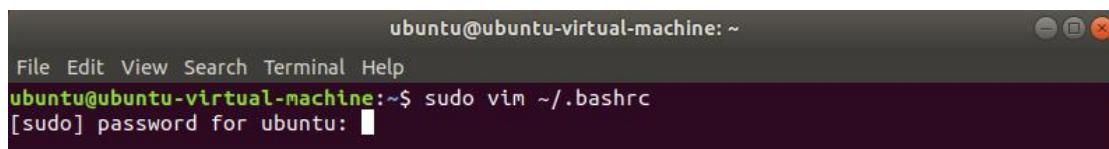
1) After connecting your mobile phone to the Wi-Fi of the Wi-Fi configuration file, enter the ArnPi FPV interface of the WonderPi and find your device in the device list.



2) Long press the device name to get the IP information and take down.



3) Open ubuntu command line terminal, enter “**sudo vim ~/.bashrc**” command and press “Enter”. If the prompt “password for ubuntu” occurs, please enter “ubuntu” and press “Enter”.



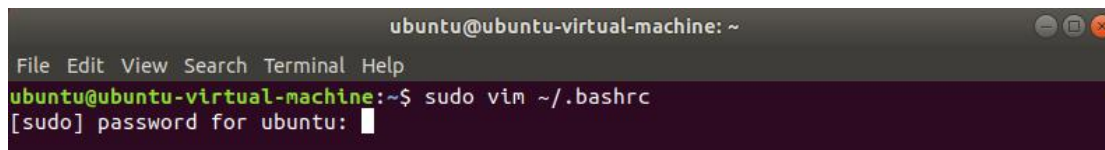
4) In the network configuration file, please change the IP of “**ROS_MASTER_URI**” to the IP we have taken down in step 2. Then press “Ese” to enter “: **wq**” and save and quit.

```
# You may want to put all your additions into a separate file like
# ~/.bash_aliases, instead of adding them here directly.
# See /usr/share/doc/bash-doc/examples in the bash-doc package.

if [ -f ~/.bash_aliases ]; then
    . ~/.bash_aliases
fi

# enable programmable completion features (you don't need to enable
# this, if it's already enabled in /etc/bash.bashrc and /etc/profile
# sources /etc/bash.bashrc).
if ! shopt -oq posix; then
    if [ -f /usr/share/bash-completion/bash_completion ]; then
        . /usr/share/bash-completion/bash_completion
    elif [ -f /etc/bash_completion ]; then
        . /etc/bash_completion
    fi
fi
source /opt/ros/melodic/setup.bash
source /home/ubuntu/arm_pi_fpv/devel/setup.bash
export ROS_MASTER_URI=http://192.168.77.160:11311
export ROS_HOSTNAME=ubuntu-virtual-machine
:wq
```

5) Please return to ubuntu command line interface and enter “**source .bashrc**” command to make the revised part effective. Now, the Raspberry Pi device and ubuntu terminal are connected to the same LAN.

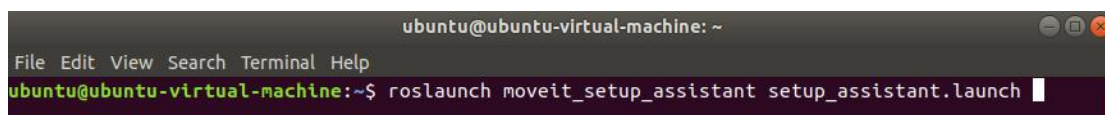


```
ubuntu@ubuntu-virtual-machine: ~
File Edit View Search Terminal Help
ubuntu@ubuntu-virtual-machine:~$ sudo vim ~/.bashrc
[sudo] password for ubuntu:
ubuntu@ubuntu-virtual-machine:~$ source .bashrc
```

2.4 MoveIt visual configuration

1) Firstly, please enter “**roslaunch**

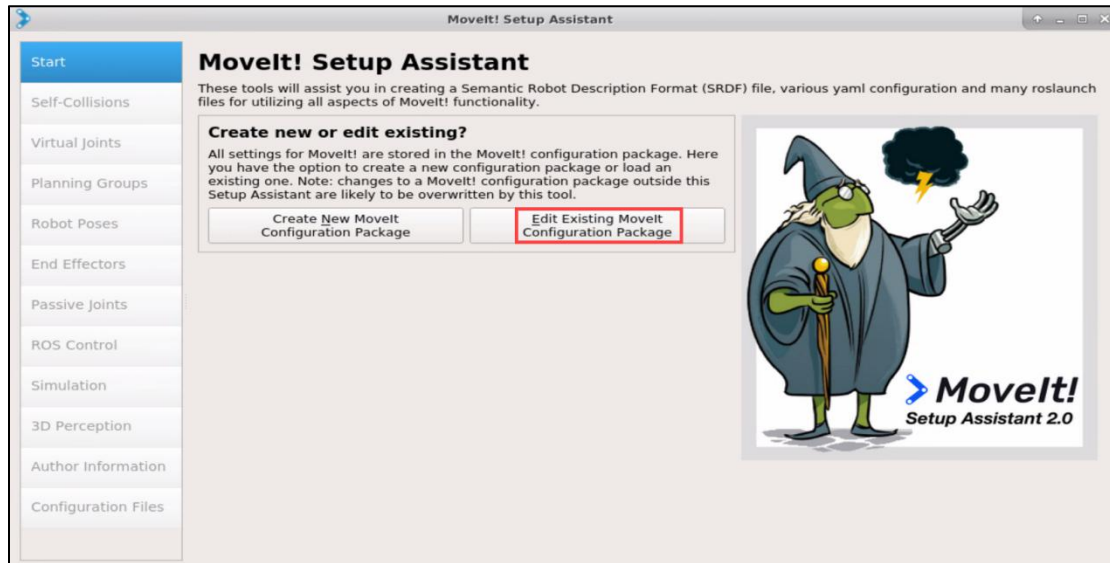
moveit_setup_assistant setup_assistant.launch” in the opened ubuntu command line terminal to start the MoveIt settings assistant.



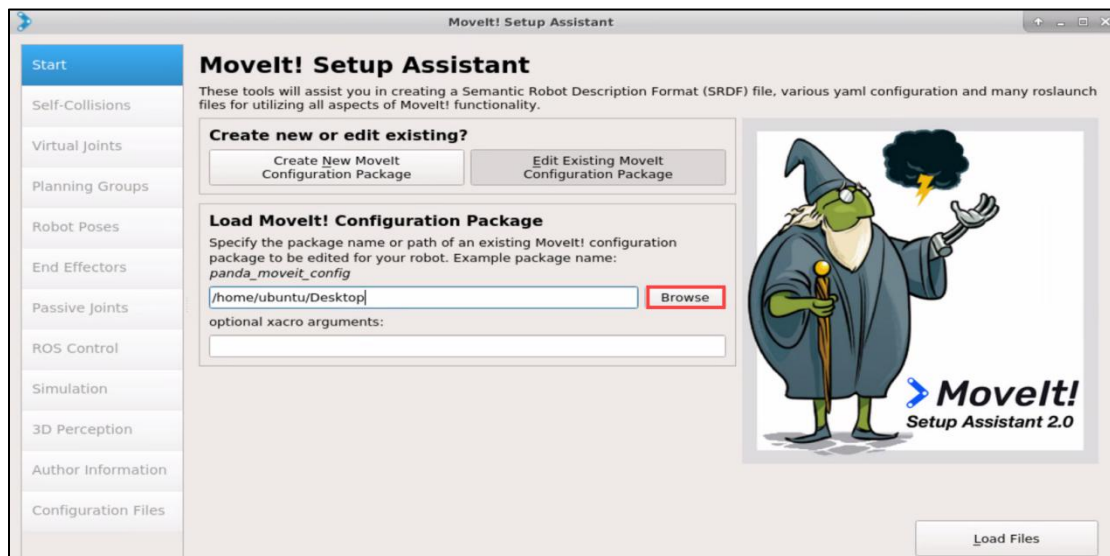
```
ubuntu@ubuntu-virtual-machine: ~
File Edit View Search Terminal Help
ubuntu@ubuntu-virtual-machine:~$ roslaunch moveit_setup_assistant setup_assistant.launch
```

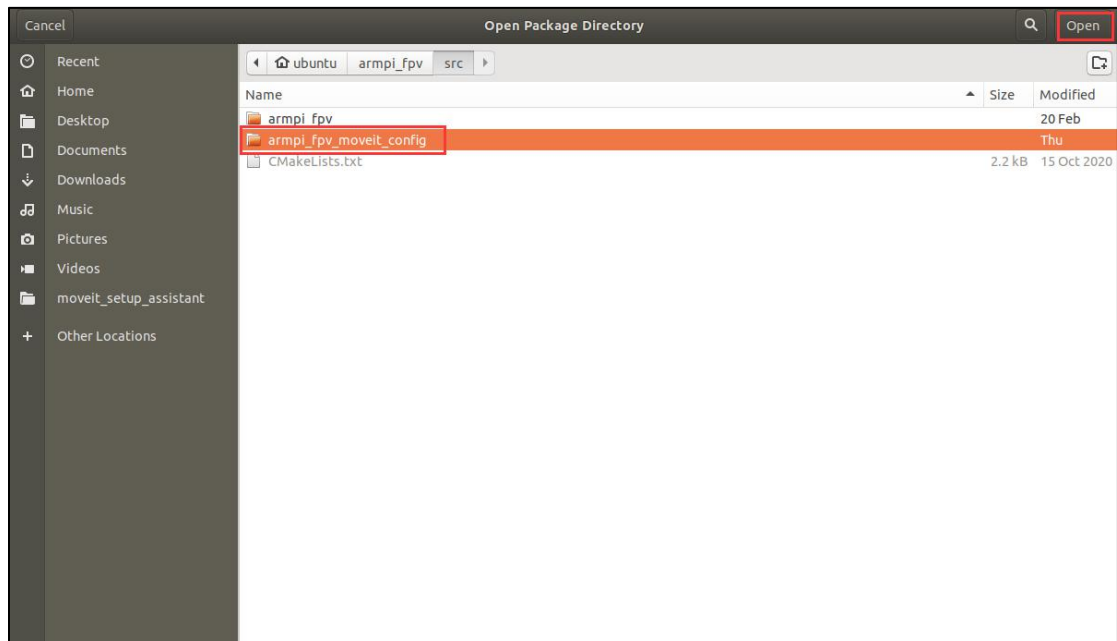
2) After successful start, we need to click “**Edit Existing MoveIt Configuration Package**”. In this instruction, we will take the existed MoveIt configuration package for example. If you want to select “**Create New MoveIt Configuration Package**”, you can move to our website to learn how to

operate.

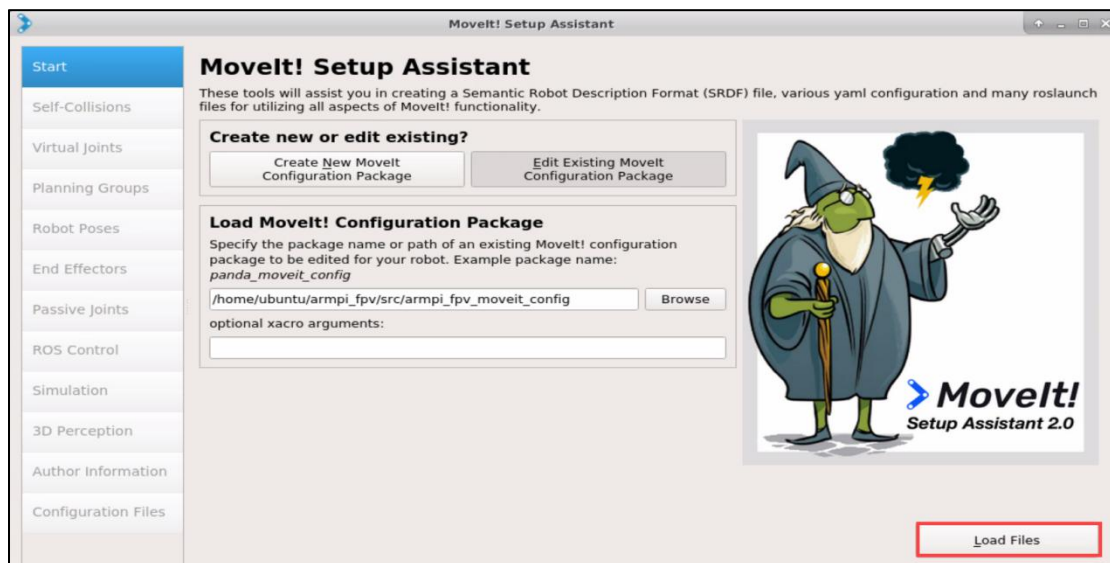


3) Next, click “Browse”, select “armpi_fpv_moveit_config” file under the path of “/home/ubuntu/armpi_fpv/src/” and click “Open”

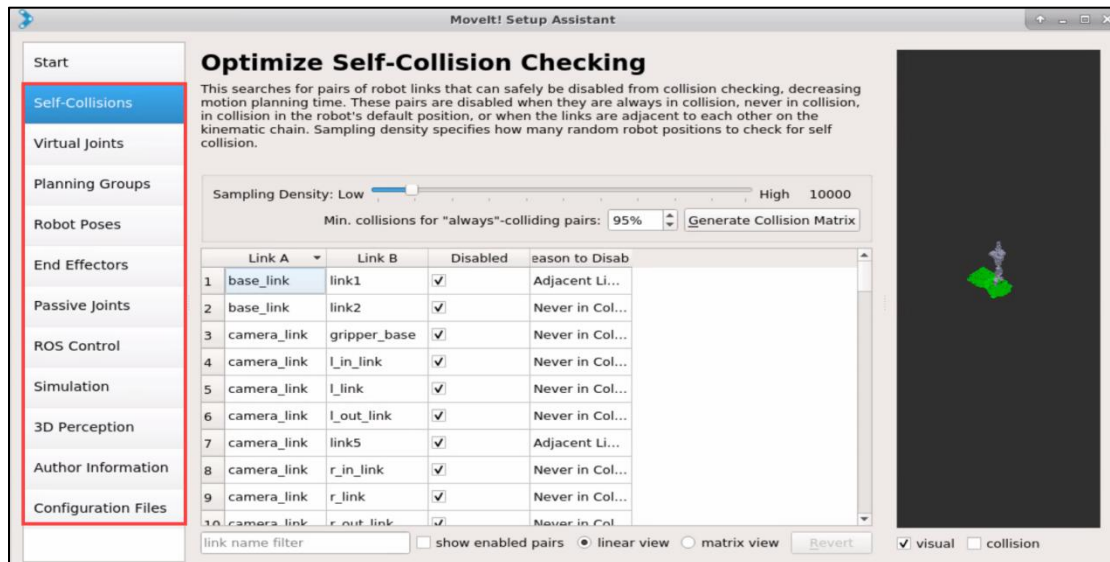




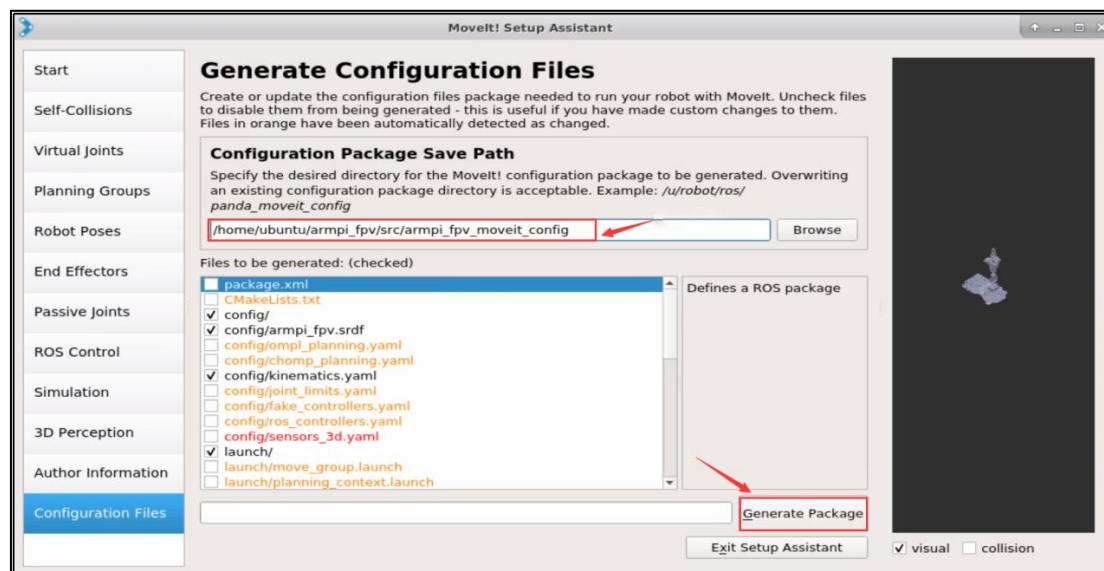
4) Then click “Load File” and wait for the file loading to 100%.



5) When the file completes loading, we can edit the content of Virtual Joints, Robot Poses, etc.



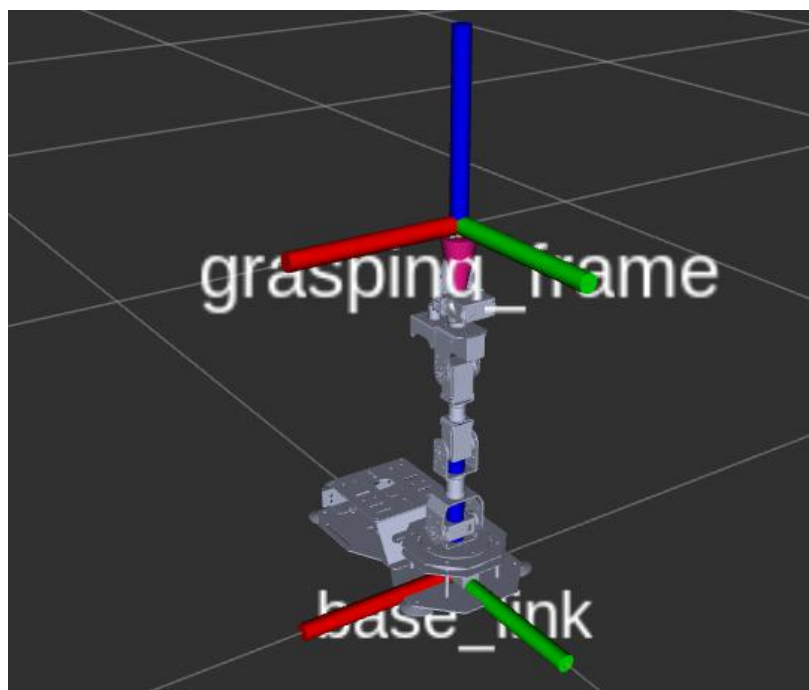
6) Having finished editing, you should click “Configuration Files”. If the prompt “Files Modified” occurs, click “OK”. Set “/home/ubuntu/armpi_fpv/src/armpi_fpv_moveit_config” as the path of configuration package and click “Generate Package”.



3. Simulated Operation

3.1 Start the ArmPi FPV kinematic model

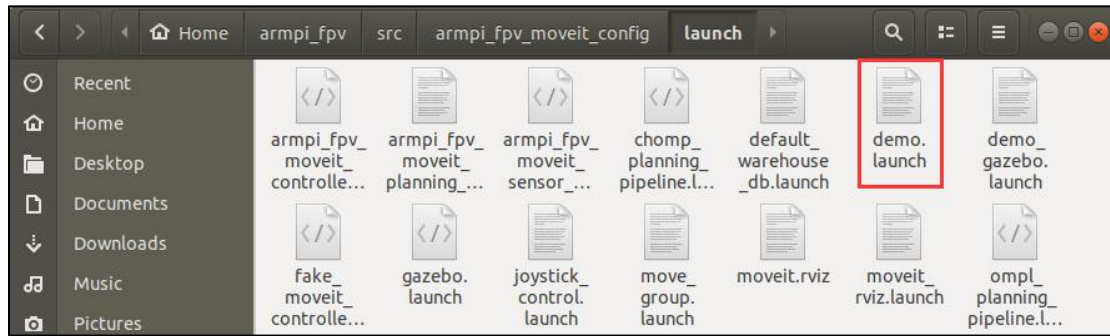
Open the Ubuntu command line terminal in the VM software and enter the command `"roslaunch armpi_fpv_moveit_config demo.launch"`. Before starting the model, please complete all the steps of 2. Set Up Environment. The finished kinematic model is as follow.



Note: The coordinate system in the figure is the base coordinate system, base_link, and the tool coordinate system, grasping_frame.

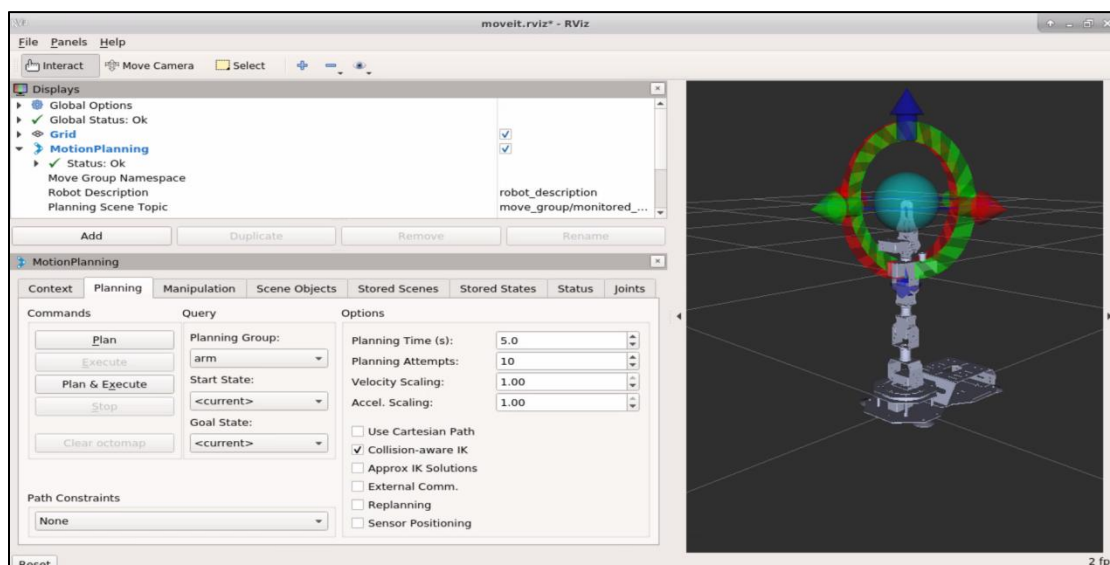
3.2 ArmPi FPV kinematic model

The kinematic model is built and you can check the details following the path: `/home/armpi_fpv/SRC/armpi_fpv_moveit_config_launch`. For the learning of kinematics model, please refer to MoveIt's official website or search on the Internet.

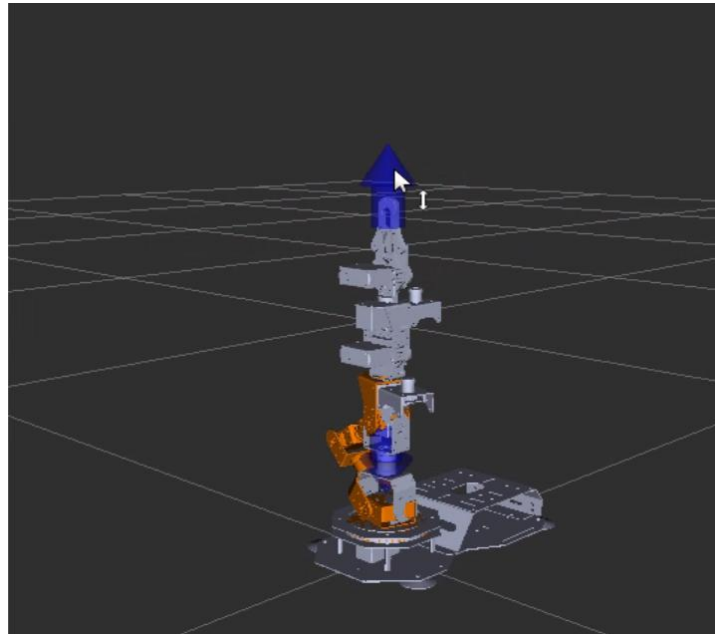


3.3 ArmPi Motion Planning realization

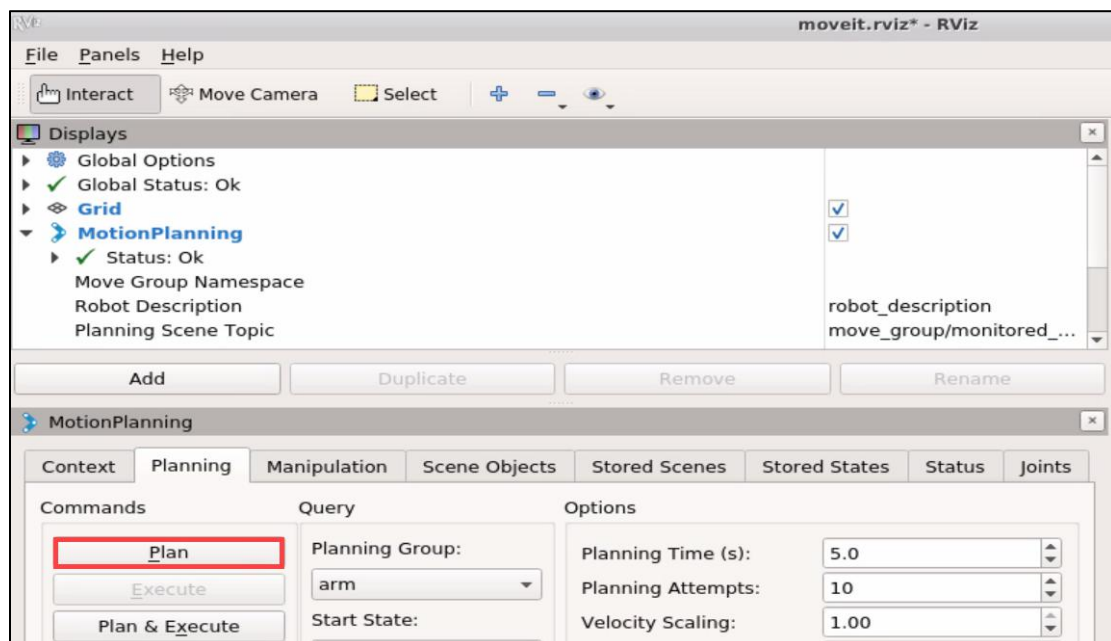
1) Firstly, start the kinematic model according to “3.1 Start the ArmPi FPV kinematic model”. The following interface will show up when you successfully start the kinematic model.



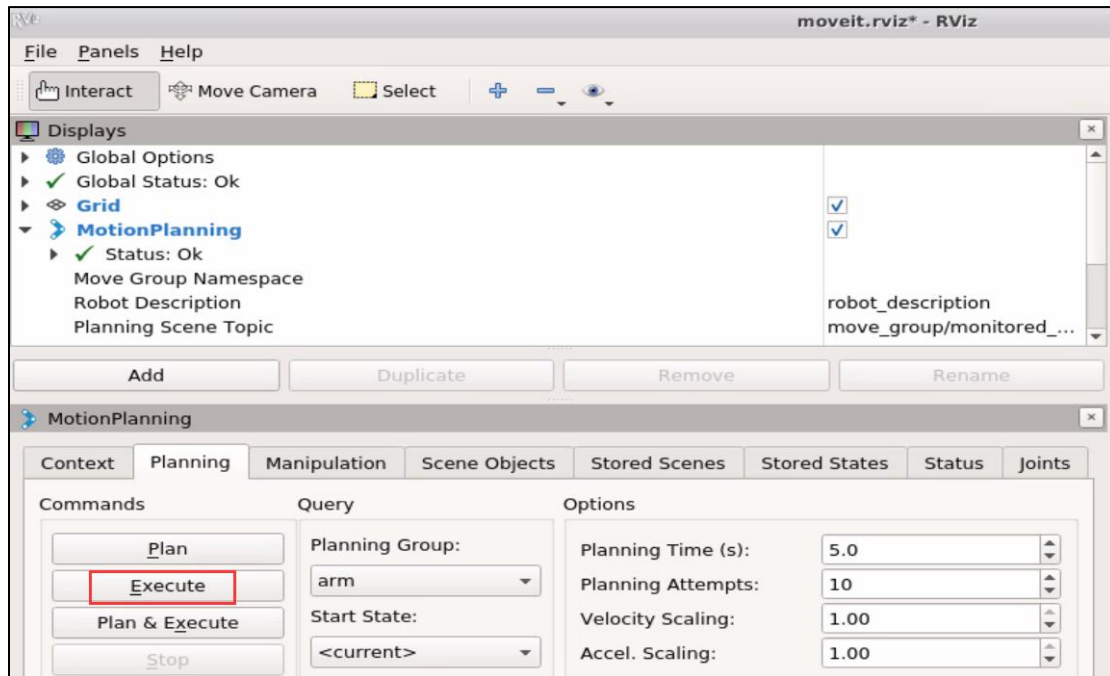
2) Next, plan for motion by directly dragging the ArmPi FPV robotic arm in the 3D space of RViz. For example, left click and hold the blue three-dimensional arrow, z axis, and drag the robotic arm down. After the dragging is completed, the motion model will move downward.



3) Then, click "Plan" under the "Planning" TAB of "MotionPlannig". And wait for the motion planning to be completed.



4) Lastly, click “Execute” under the "Planning" TAB of "MotionPlannig" and execute the programmed motion.



Note: For further study of MoveIt, please move to MoveIt's website, <https://moveit.ros.org/>.